

Creative Full-Stack & AI Development Internship

Internship Duration

60 Days

This internship is designed for students interested in working on **Full-Stack Development, AI Integration, System Design, and Creative Product Development**.

Candidates will work on one of the given problem statements and are expected to develop a **complete working application**, including frontend, backend, database, AI integration, and deployment.

The objective is not to develop only an AI model, Python script, or prototype. The final outcome should be a **functional full-stack product**.

Expected Technology Stack

Frontend (java script)

- React.js
- JavaScript / TypeScript
- Tailwind CSS

Backend (python/java script)

Preferred:

- Python
- FastAPI or node js

AI / ML Technologies

Based on the selected problem statement, candidates may work with technologies such as:

- Python AI/ML Libraries, Computer Vision, Large Language Models (LLMs), Speech-to-Text Models, Natural Language Processing (NLP), Recommendation Systems, Embedding Models

Prior experience in all of the above technologies is not mandatory. Candidates who do not have exposure to some of these areas will have the opportunity to **learn, explore, and implement them during the 60-day internship period**.

We primarily expect candidates to have a strong interest in **learning new technologies, problem-solving, experimentation, and building practical AI-powered applications**.

Problem Statement 01

AI-Powered Face Recognition & Smart Gallery

Develop a full-stack **AI-powered face recognition and photo gallery application**.

The system should be capable of analyzing uploaded images, detecting faces, identifying or grouping similar individuals, and automatically organizing photographs based on the people appearing in them.

The application can include features such as:

- Face detection
- Face recognition
- Face similarity matching
- Person-wise photo grouping
- Smart photo gallery
- Search by person
- Unknown face identification
- Naming identified people
- Multiple-face detection
- Image management

The candidate should focus on both the **AI/ML implementation and creative user experience of the gallery application**.

The final solution should include a frontend, backend APIs, database, image storage, and AI-based face processing.

Problem Statement 02

AI-Based Fraud Detection from Customer Conversations

Develop a full-stack **AI-powered conversation analysis and fraud-risk detection system**.

The application will receive customer or claim-related voice conversations that may be in:

- Tamil
- English
- Tamil-English mixed conversations

The system should convert the audio into text and analyze the conversation to identify potential suspicious information, inconsistencies, contradictions, or fraud-risk indicators.

Possible features include:

- Audio upload
- Speech-to-Text

- Tamil and English conversation support
- Conversation transcription
- Conversation summarization
- Key information extraction
- Contradiction detection
- Suspicious statement identification
- Fraud-risk indicators
- Risk score
- Highlighted transcript sections
- Reviewer dashboard

The application should assist human reviewers by identifying **possible fraud indicators**, rather than automatically declaring a customer fraudulent.

The final solution should be developed as a complete full-stack investigation and analysis application.

Problem Statement 03

Domain-Specific AI Application

Select a specific business domain, identify its real-world problems, and develop a **creative AI-powered full-stack solution** for that domain.

Example domains include:

- Clinic
- Restaurant
- Education
- Real Estate
- Logistics
- Travel
- Hospitality
- Retail
- Fitness
- Recruitment

Candidates should first understand the selected industry's workflow and identify genuine customer or operational pain points.

Based on this analysis, candidates should design and implement an appropriate AI solution.

For example:

Clinic

Possible solutions:

- AI appointment assistant
- Doctor/service recommendation
- Patient query assistant
- Follow-up management

Restaurant

Possible solutions:

- AI menu recommendations
- Food suggestions
- Dietary preference recommendations
- Smart ordering
- Reservation assistant
- Customer support

Candidates are encouraged to explore their own industry and develop a **unique and practical solution rather than simply adding a chatbot.**

The final application should demonstrate creativity, system design, AI integration, UI/UX, backend development, and database integration.

Problem Statement 04

AI-Powered E-Commerce System

Design and develop a **full-stack AI-powered e-commerce application.**

Candidates should study the e-commerce customer journey and identify areas where Artificial Intelligence can improve the shopping experience.

Possible AI features include:

- Personalized product recommendations
- AI product search
- Product suggestions
- Gift recommendations
- Product comparison
- AI shopping assistant
- Smart product bundles
- Personalized home page
- Review summarization
- Visual product search
- Smart filters
- Product recommendation based on user interests

For example, a user could ask:

"Suggest a birthday gift under ₹2,000 for someone interested in technology."

The AI system should analyze the requirement and provide relevant product recommendations.

The application should also contain basic e-commerce functionality such as:

- Product listing
- Product details

- Categories
- Search
- Cart
- User account
- Checkout flow
- Order management

The objective is to design an **AI-integrated e-commerce system**, rather than simply developing a normal e-commerce website with an additional chatbot.

Internship Expectations

During the **60-day internship**, candidates are expected to:

- Understand the assigned problem statement
- Research possible solutions
- Design the application architecture
- Document the project

The final submission should demonstrate:

Creative Thinking + Full-Stack Development + AI Integration + System Design

Candidates are encouraged to explore different approaches and technologies while maintaining a structured, scalable, and practical application architecture.